Name: *In the Name of God, the Most High*

STN:

Deep Learning
Computer Assignment 4

Using Recurrent Neural Networks to Generate Verses of the Shahnameh

In this exercise, we aim to train a recurrent neural network using the textual data of the Shahnameh. Ultimately, given a sequence from the Shahnameh, the network should be able to continue it on its own and generate poetry in the same meter.

1) In a deep learning project, the most important part is creating and collecting the dataset. The dataset provided to you is in raw form. First, prepare the data with appropriate reasoning and explanation, and then use it.

2) Create a suitable model for your data, compile the model with an appropriate loss function, and train it for a suitable number of iterations (epochs).

3) Generate, for example, a 1000-character sequence using the trained network and evaluate how successful the network you trained has been.

Additional Notes:

1) Starter code has been provided. If you have difficulty running it on your own system, use the Google Colab environment.

2) For the Persian explanations and the report, there is no need to provide a separate PDF file; write the Persian explanations in the same .ipynb file.

3) Use deep learning libraries such as TensorFlow and Keras to complete this exercise.

4) To increase execution speed, you may select a GPU as the processor type.

5) This is an individual assignment, and copying is not permitted. However, you may consult others regarding their experience writing code and use discussion for analysis and comparison.

Good luck!